

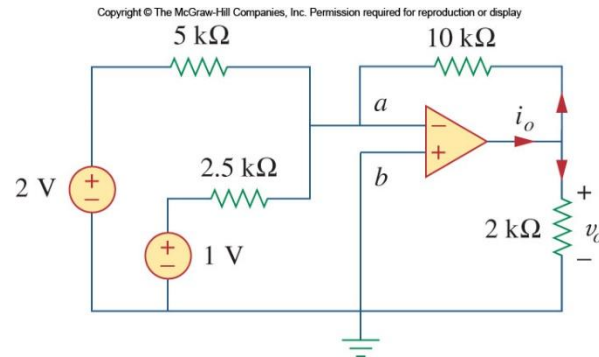
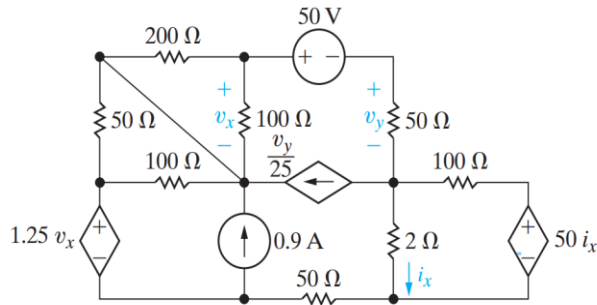
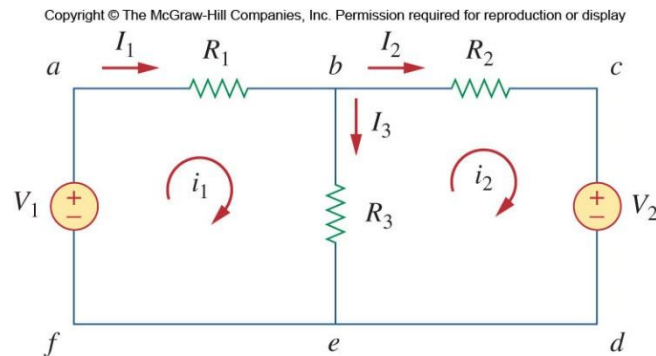
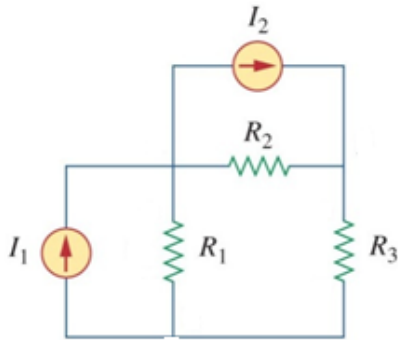


Lecture 6

- RC/RL First-Order Circuits

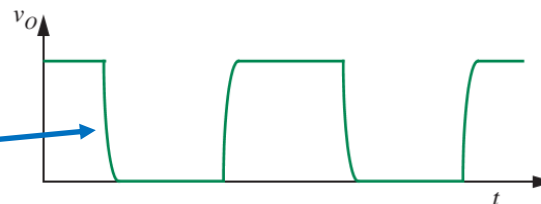
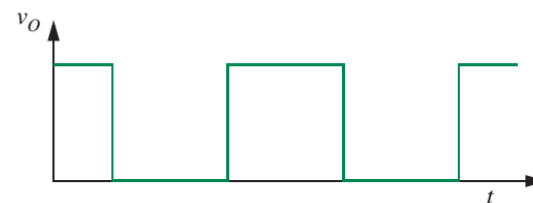
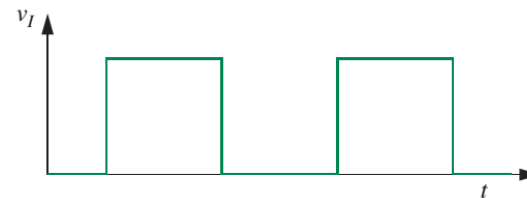
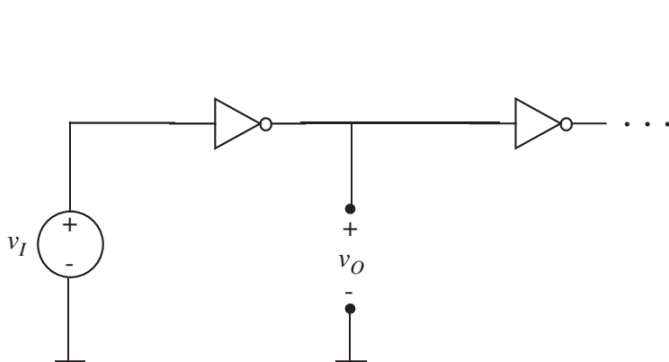
Temporal Behavior of Circuit Responses

- Till now we discussed static analysis of a circuit
 - Responses at a given time depend only on inputs at that time.
 - Circuit responds to input changes infinitely fast.



Temporal Behavior of Circuit Responses

- From now on we start to discuss dynamic circuit
 - **Time-varying** sources and responses



We have to introduce capacitors and inductors to explain such temporal behavior.

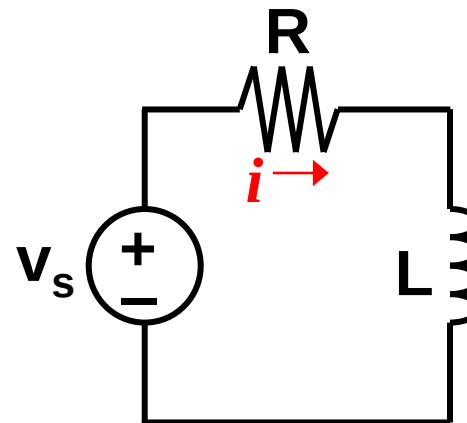
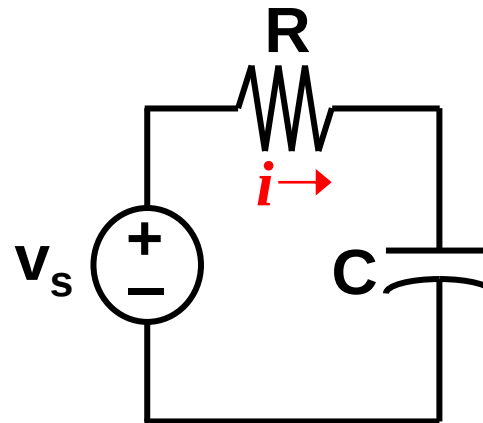


Outline

- Natural response of RC/RL circuits
- Step response of RC/RL circuits

RC and RL Circuits

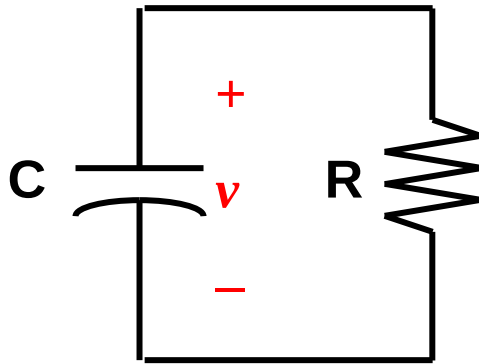
- A circuit that contains only sources, resistors and a capacitor is called an **RC circuit**.
- A circuit that contains only sources, resistors and an inductor is called an **RL circuit**.





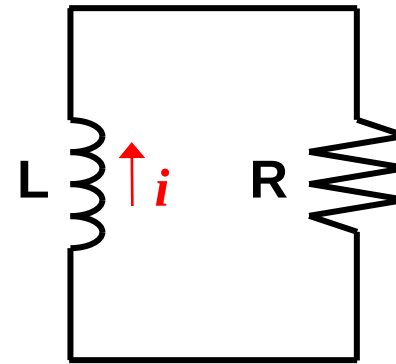
RC and RL Circuits

RC Circuit



- Capacitor voltage cannot change instantaneously
- In steady state, a capacitor behaves like an open circuit

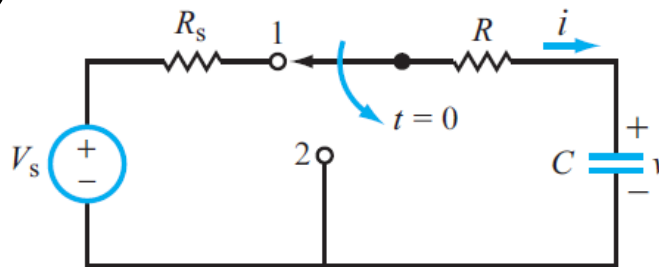
RL Circuit



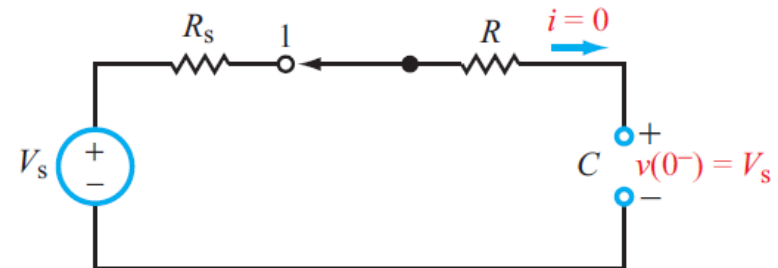
- Inductor current cannot change instantaneously
- In steady state, an inductor behaves like a short circuit.

Natural Response of a Charged Capacitor

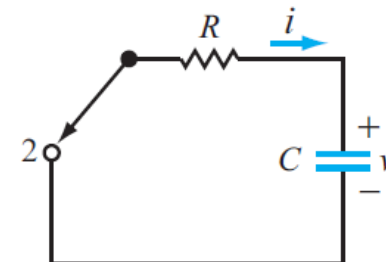
Behavior (*i.e.*, current and voltage) when stored energy in the inductor or capacitor is released to the resistive part of the network (containing no independent sources).



(a) $t = 0^-$ is the instant just before the switch is moved from terminal 1 to terminal 2;

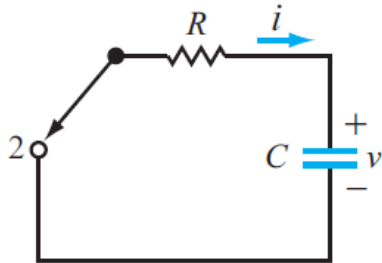


(b) $t = 0$ is the instant just after it was moved, $t = 0$ is synonymous with $t = 0^+$.

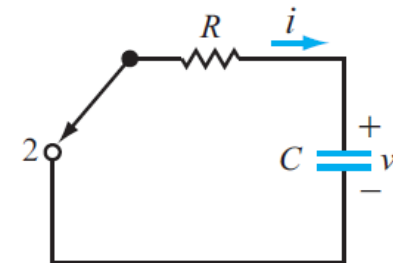




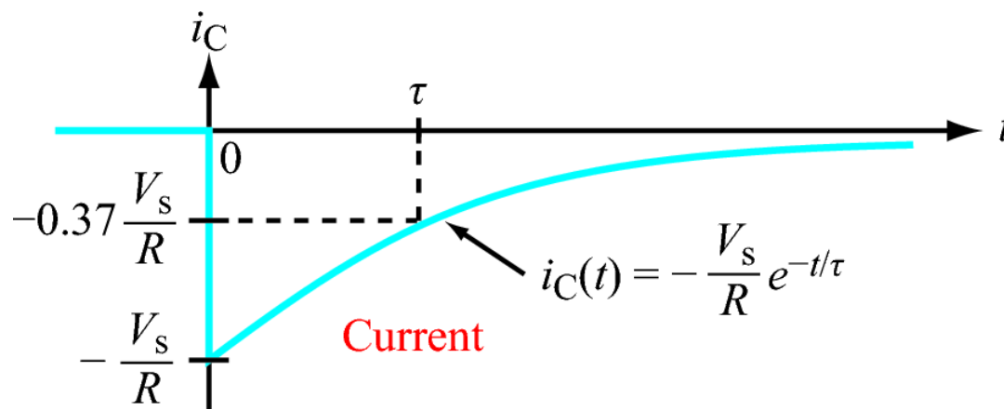
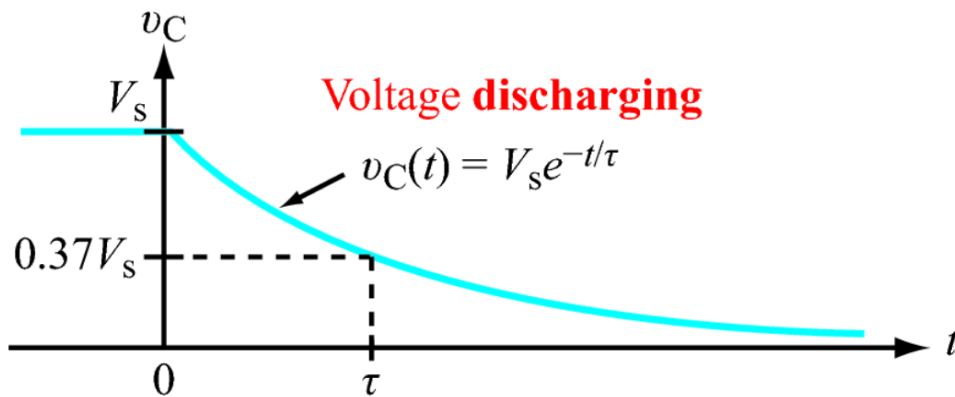
Natural Response of a Charged Capacitor



Natural Response of RC

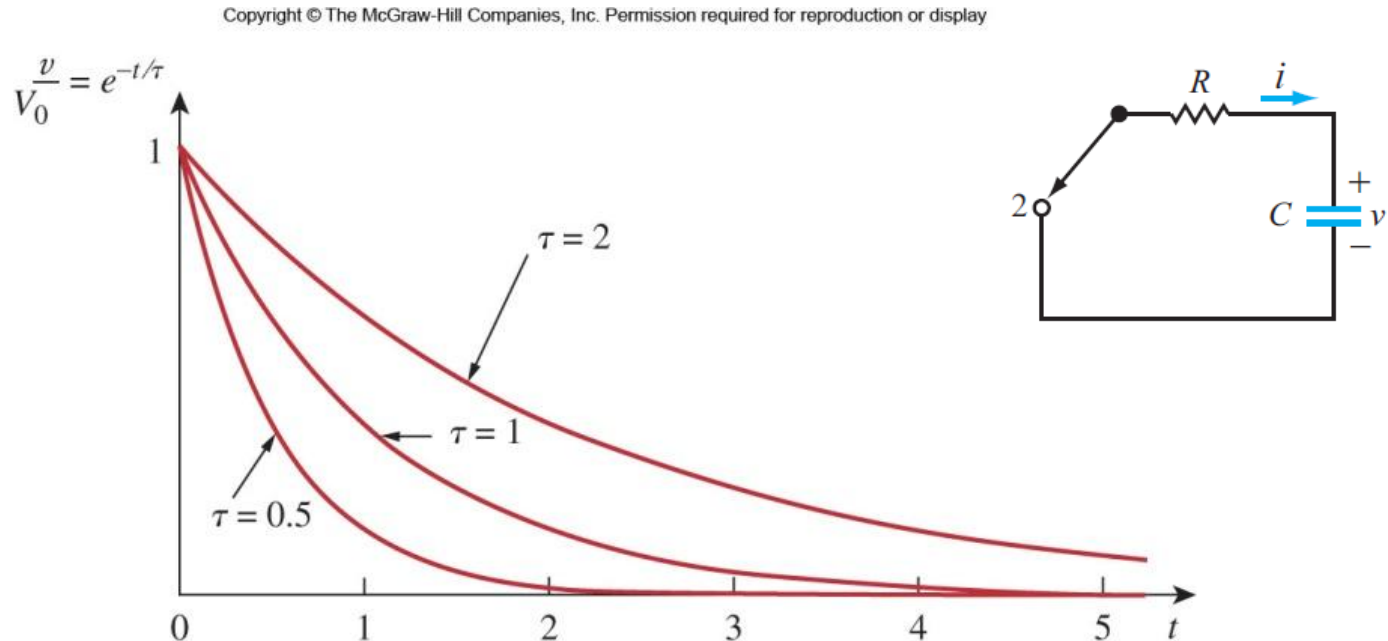


Time constant: $\tau = RC$



Time Constant $\tau (= RC)$

- A circuit with a small time constant has a fast response and vice versa.

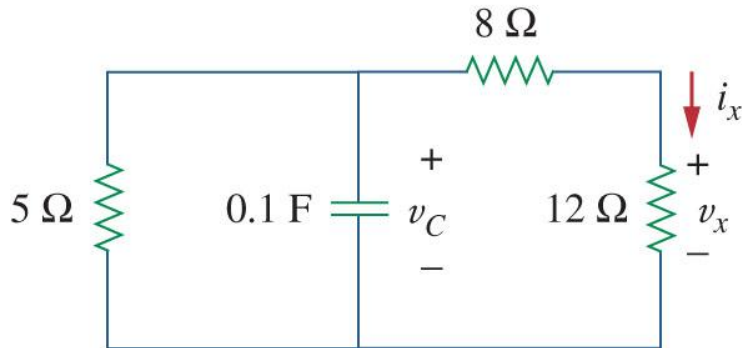




Example

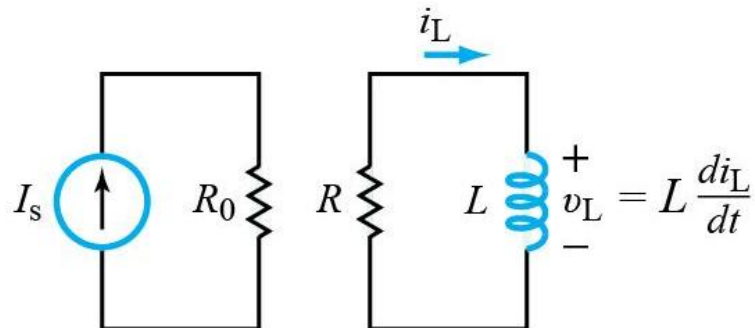
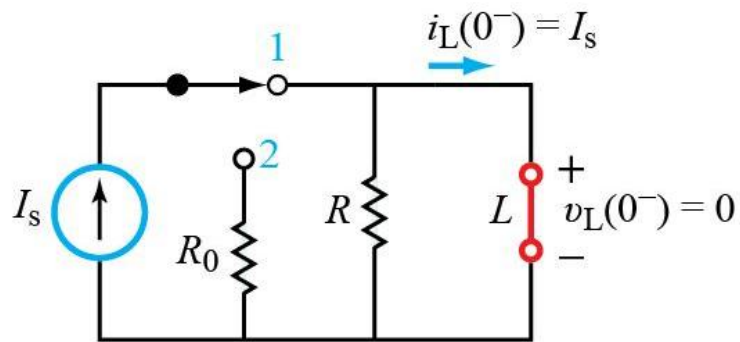
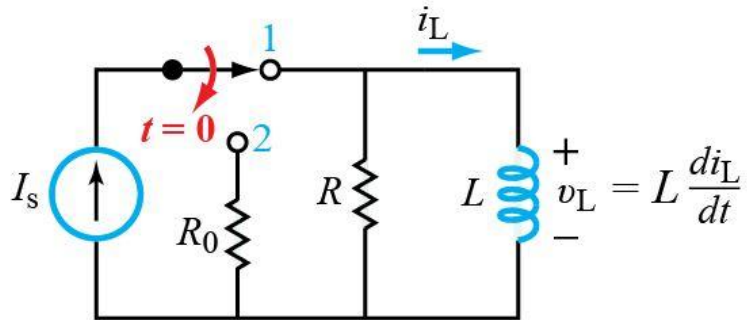
- In the circuit below, let $v_C(0) = 15\text{V}$. Find v_C , v_x , and i_x for $t > 0$.

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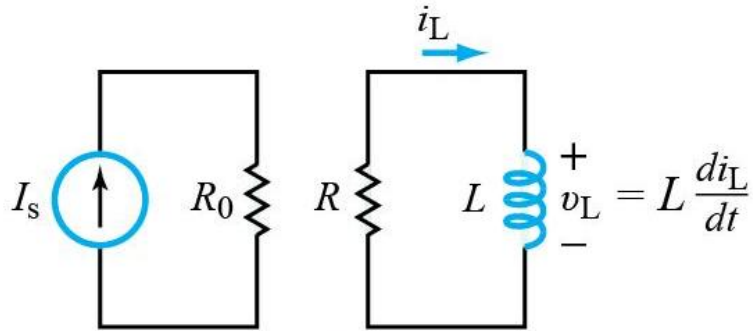


Natural Response of the RL Circuit



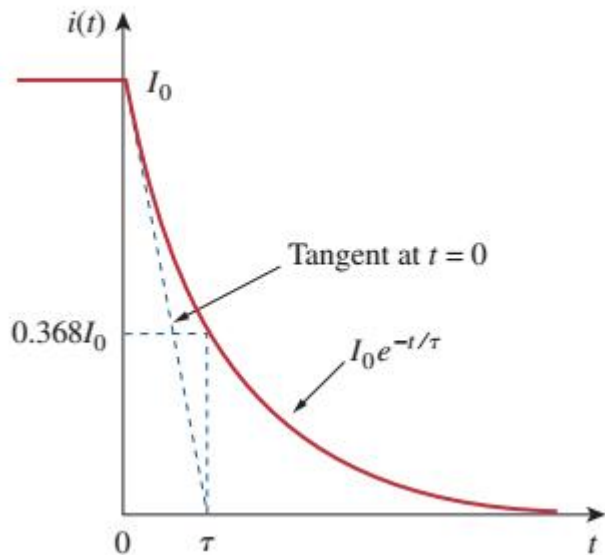


Natural Response of the RL Circuit





Natural Response of the RL Circuit

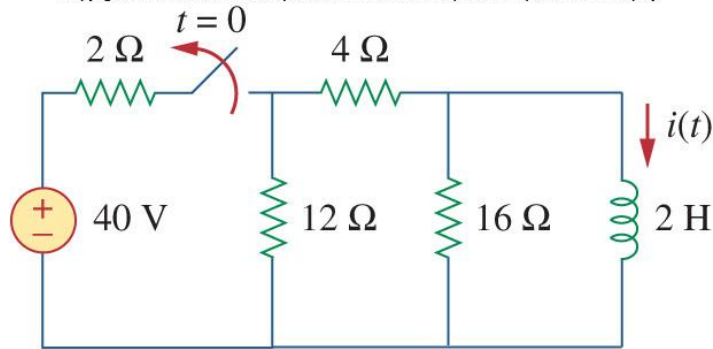




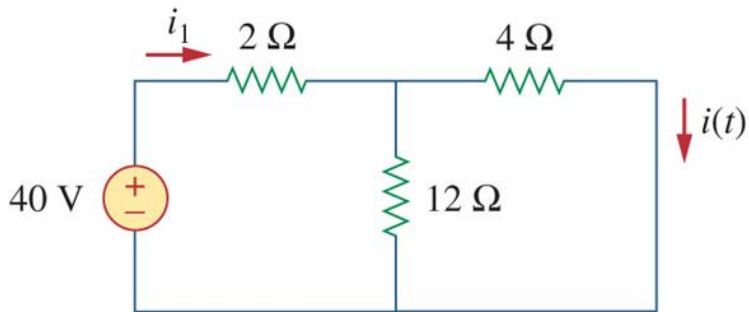
Example

- The switch in the circuit below has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

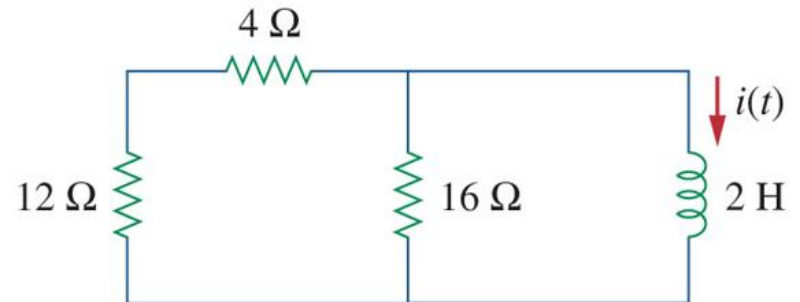
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When $t < 0$

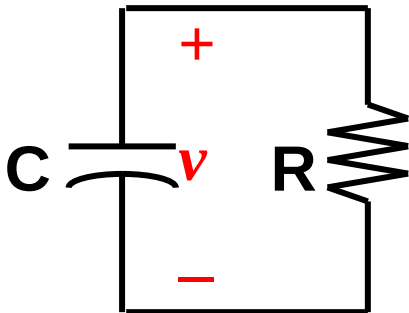


When $t > 0$



Natural Response Summary

RC Circuit



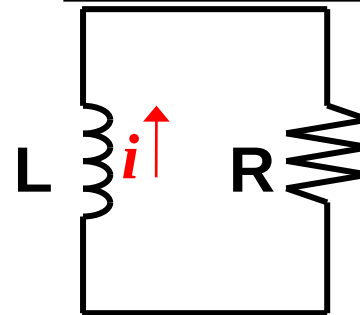
- Capacitor voltage cannot change instantaneously

$$v(0^-) = v(0^+)$$

$$v(t) = v(0)e^{-t/\tau}$$

- time constant $\tau = RC$

RL Circuit



- Inductor current cannot change instantaneously

$$i(0^-) = i(0^+)$$

$$i(t) = i(0)e^{-t/\tau}$$

- time constant $\tau = \frac{L}{R}$



7.16 Determine the time constant for each of the circuits in Fig. 7.96.

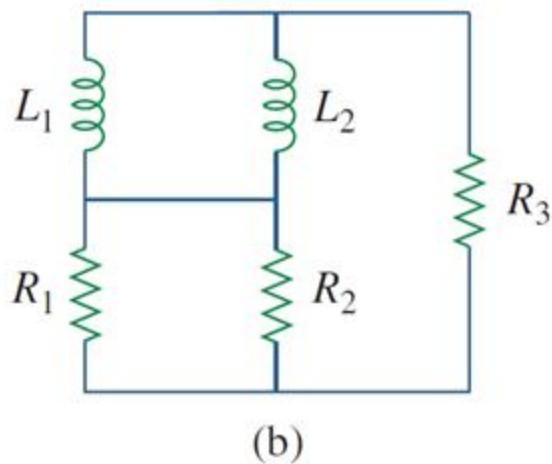
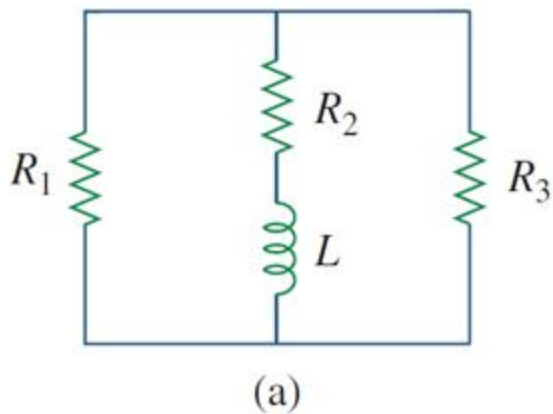


Figure 7.96
For Prob. 7.16.



7.19 In the circuit of Fig. 7.99, find $i(t)$ for $t > 0$ if $i(0) = 6$ A.

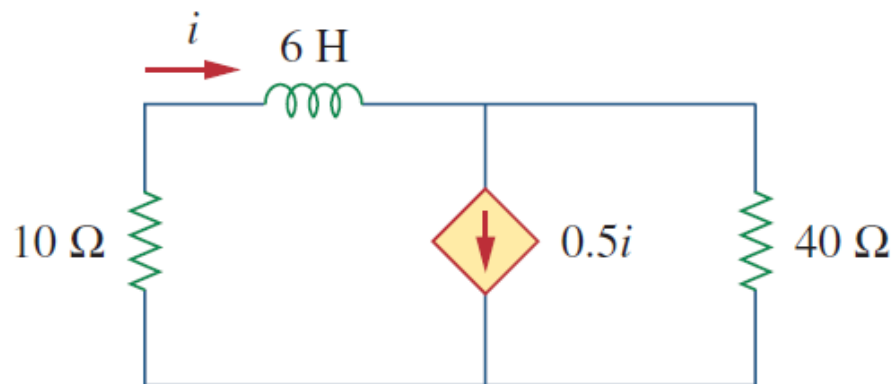
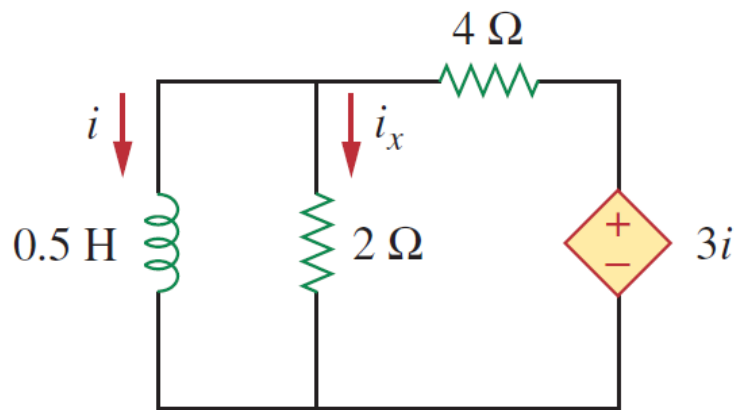


Figure 7.99
For Prob. 7.19.



Assuming that $i(0) = 10$ A, calculate $i(t)$ and $i_x(t)$





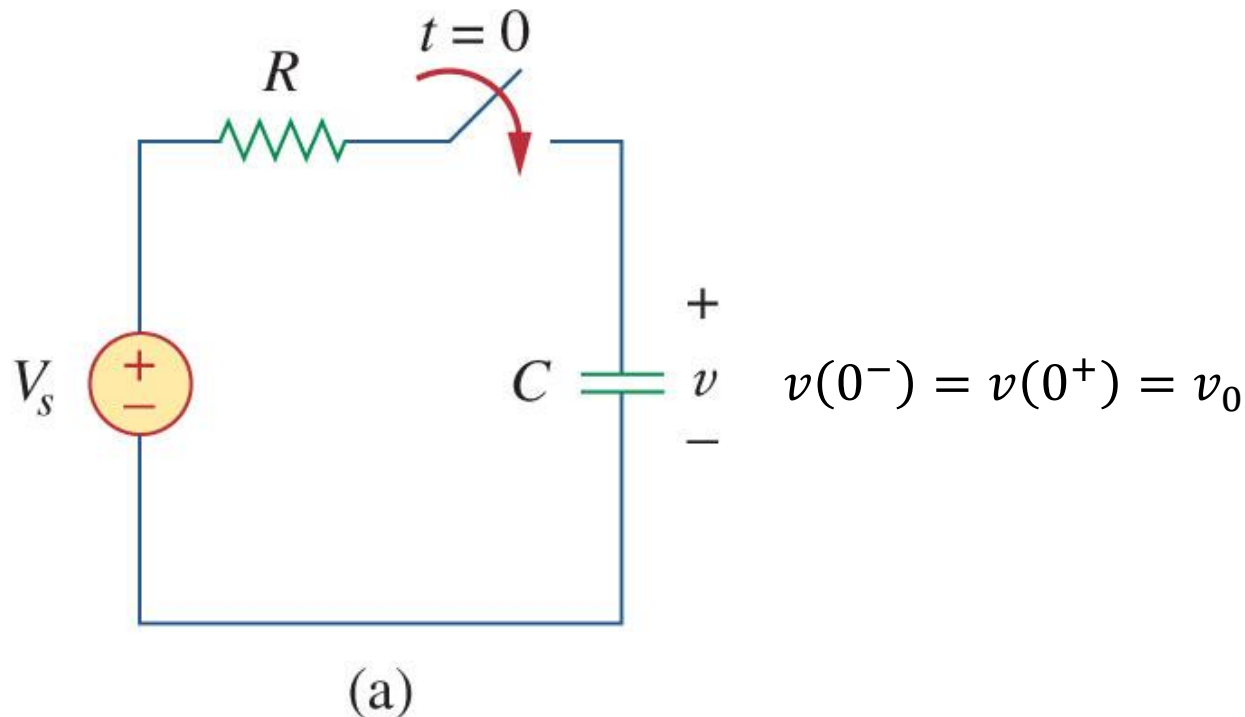
Outline

- Natural response of RC/RL circuits
- Step response of RC/RL circuits

Step Response of RC Circuit

- When a DC source is suddenly applied to a RC circuit, the circuit response is known as the step response.

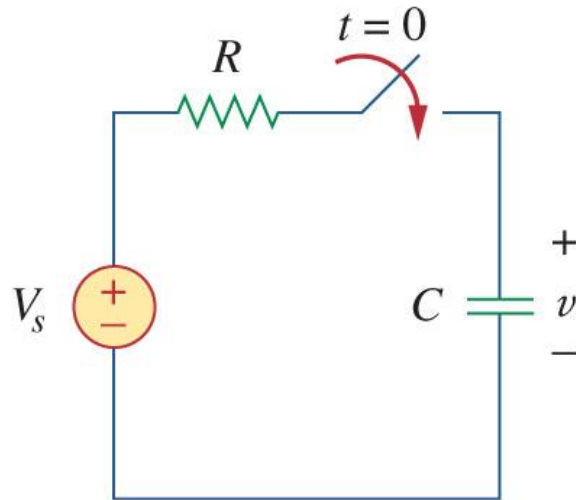
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Step Response of the RC Circuit

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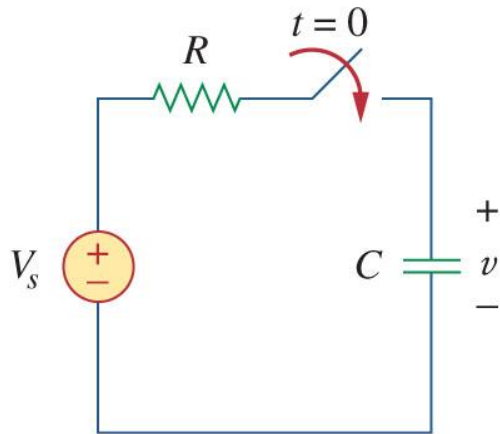


$$v(0^-) = v(0^+) = v_0$$



Step Response of the RC Circuit

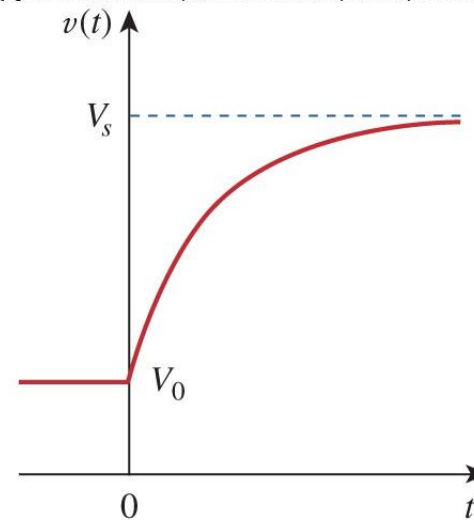
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$$v(0^-) = v(0^+) = v_0$$

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$

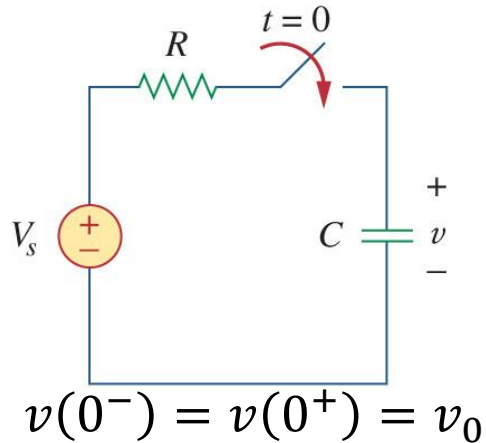
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- This is known as the complete response, or total response.

Forced Response

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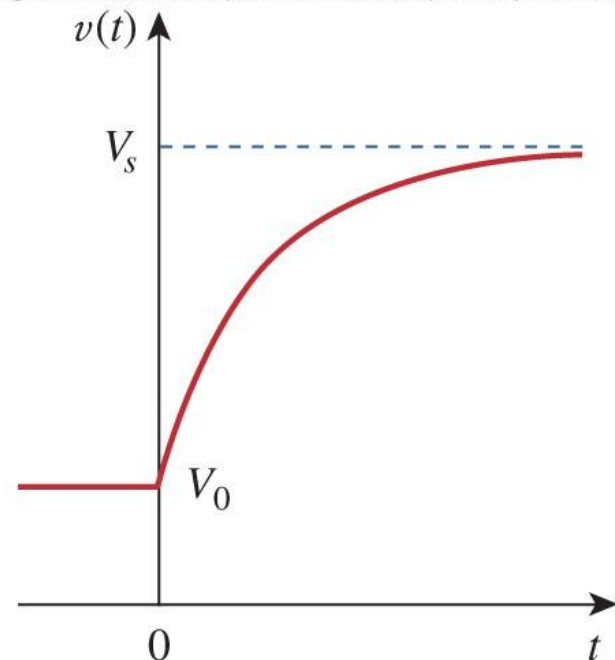
- The complete response

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$

can be written as:

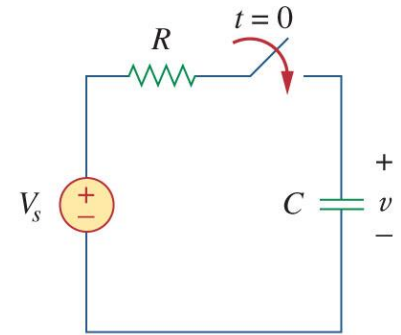
$$v = v_n + v_f$$

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Another Perspective

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$



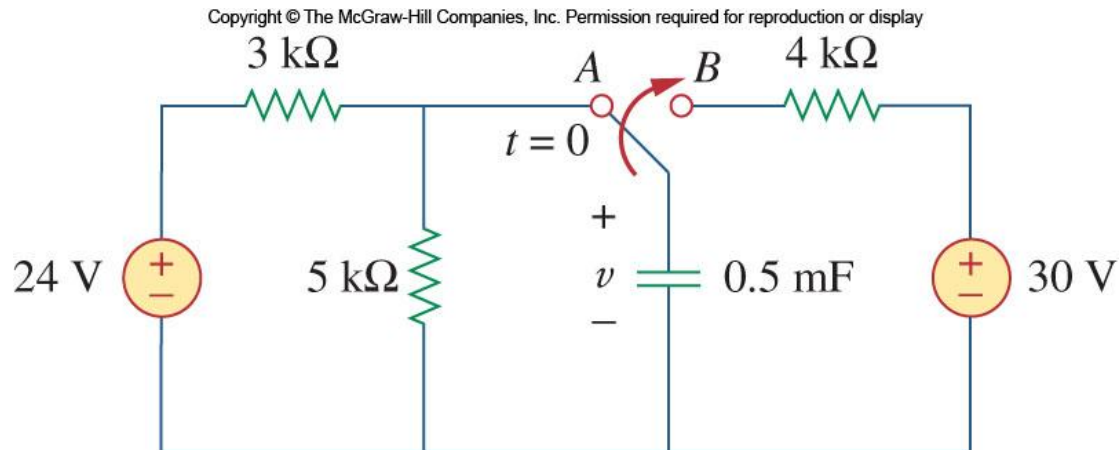
- Another way to look at the response is to break it up into the transient response and the steady state response:

$$v(t) = \underbrace{v(\infty)}_{\text{steady } v_{ss}} + \underbrace{[v(0) - v(\infty)]e^{-t/\tau}}_{\text{transient } v_t}$$



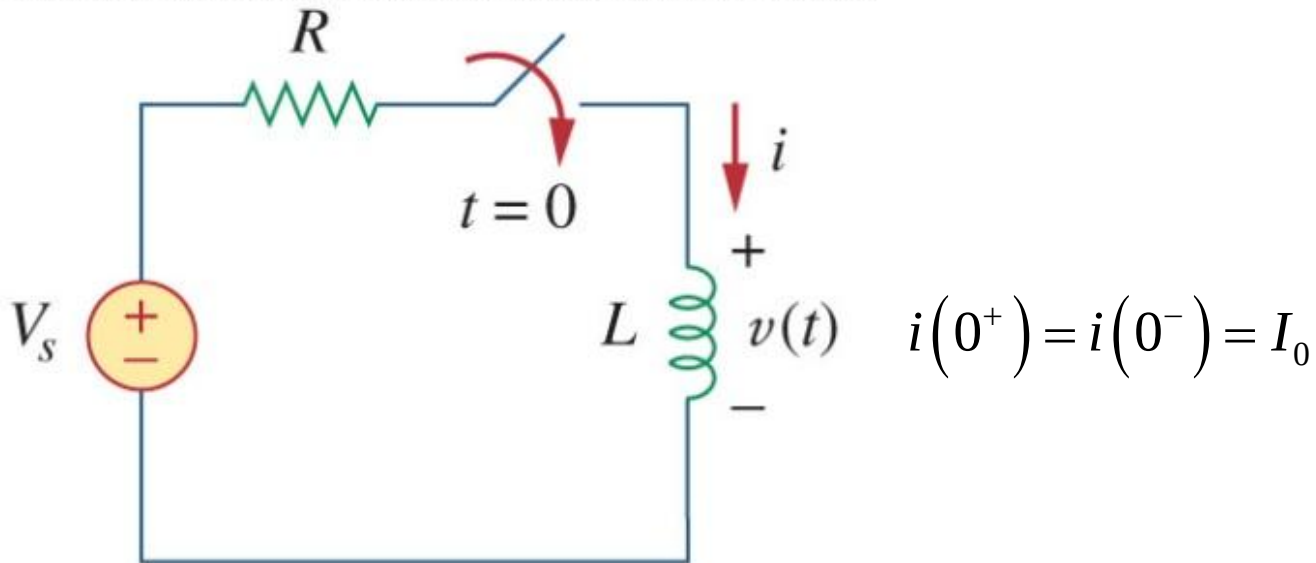
Example

- The switch has been in position A for a long time. At $t = 0$, the switch moves to B. Find $v(t)$.



Step Response of the RL Circuit

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Step Response of the RL Circuit

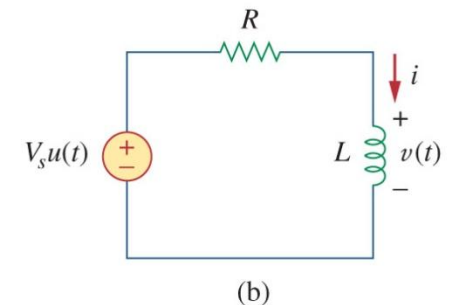
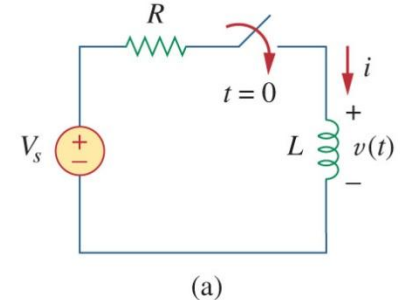
- We will use the transient and steady state response approach.
- We know that the transient response will be an exponential:

$$i_t = Ae^{-t/\tau}$$

- After a sufficiently long time, the current will reach the steady state:

$$i_{ss} = \frac{V_s}{R}$$

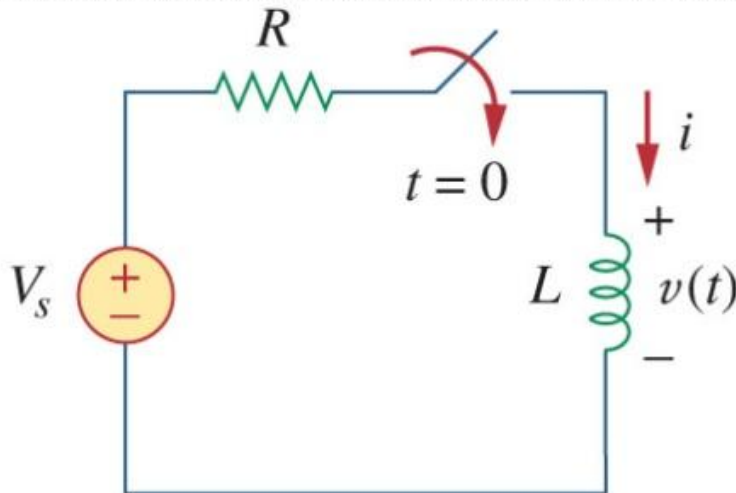
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Step Response of RL Circuit

- This yields an overall response of:

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$$i = Ae^{-t/\tau} + \frac{V_s}{R}$$

$$i(0^+) = i(0^-) = I_0 \quad A = I_0 - \frac{V_s}{R}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

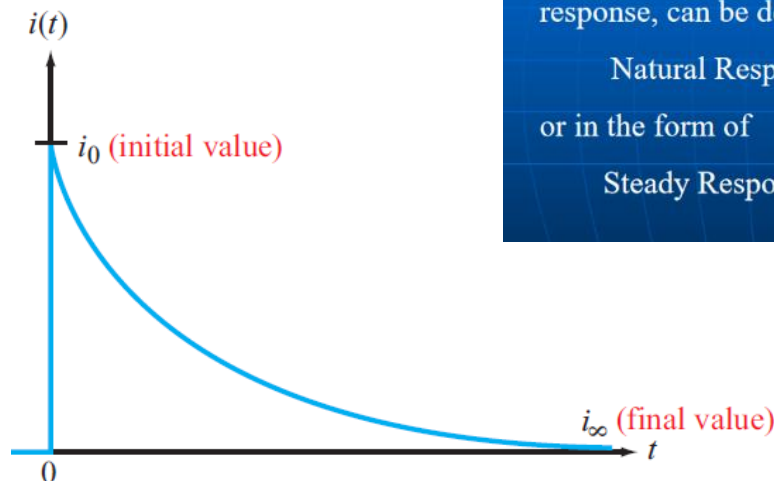
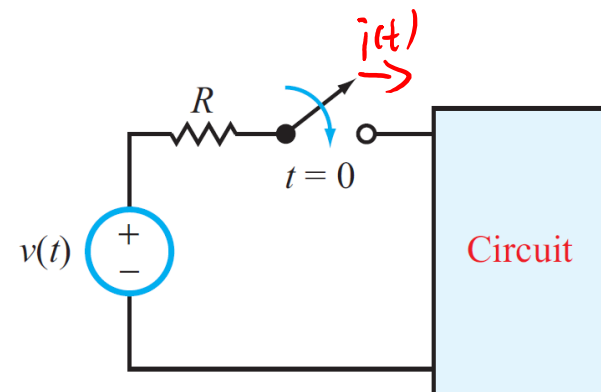
Response of a Circuit

• *Circuit (dynamic) response*

- the reaction of a certain voltage or current in the circuit to change, such as the adding of a new source, the elimination of a source, in the circuit configuration.

• *Transient response*

- Behavior when voltage or current source are **suddenly** applied to or removed from the circuit due to switching.
- Temporary behavior



The solution of a linear circuit, called dynamic response, can be decomposed into

Natural Response + Forced Response

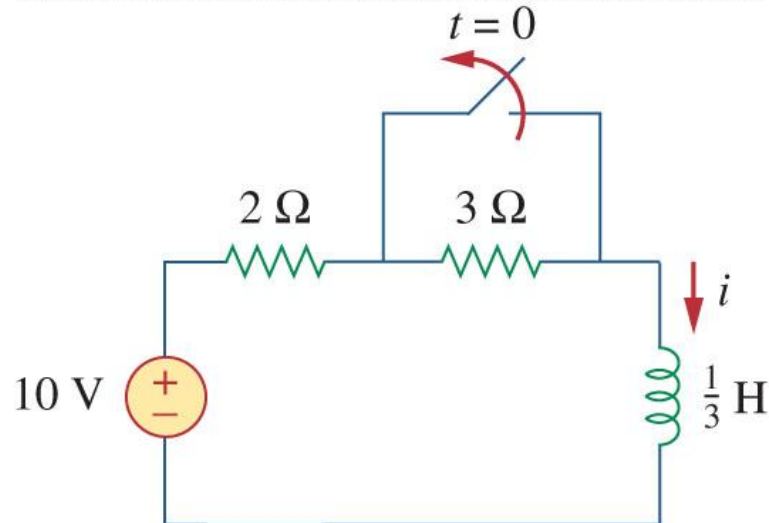
or in the form of

Steady Response + Transient Response

Example

- Find $i(t)$ in the circuit for $t > 0$. Assume that the switch has been closed for a long time.

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General Procedure for Finding RC/RL Response

1. Identify the variable of interest

- For RL circuits, it is usually the inductor current $i_L(t)$.
- For RC circuits, it is usually the capacitor voltage $v_c(t)$.

2. Determine the initial value (at $t = t_0^-$ and t_0^+) of the variable

- Recall that $i_L(t)$ and $v_c(t)$ are continuous variables:

$$i_L(t_0^+) = i_L(t_0^-) \quad \text{and} \quad v_c(t_0^+) = v_c(t_0^-)$$

- Assuming that the circuit reached steady state before t_0 : use the fact that **an inductor behaves like a short circuit in steady state** or that **a capacitor behaves like an open circuit in steady state**.



Procedure (cont'd)

3. Calculate the final value of the variable (as $t \rightarrow \infty$)

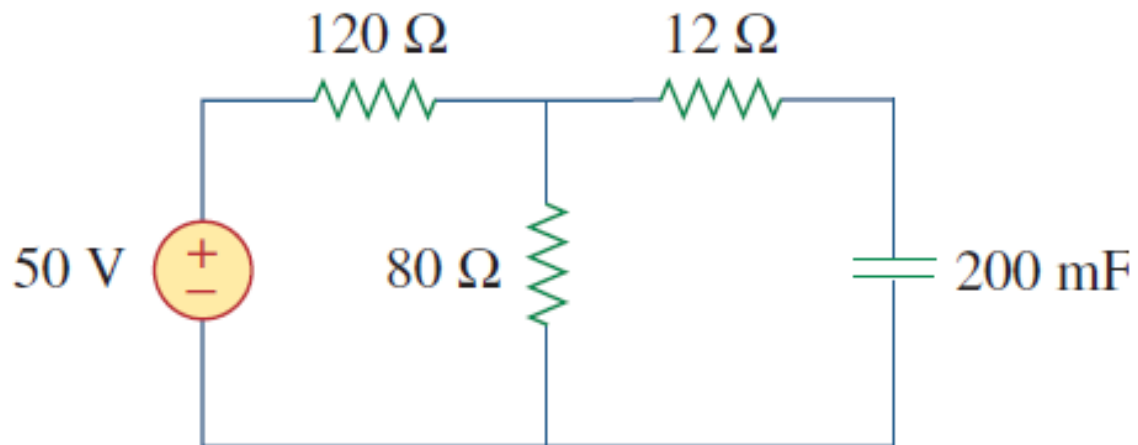
- Again, make use of the fact that **an inductor behaves like a short circuit in steady state ($t \rightarrow \infty$)** or that **a capacitor behaves like an open circuit in steady state ($t \rightarrow \infty$)**.

4. Calculate the time constant for the circuit

- **$\tau = CR$ for an RC circuit** where **R** is the Thévenin equivalent resistance “seen” by the capacitor.
- **$\tau = L/R$ for an RL circuit**, where **R** is the Thévenin equivalent resistance “seen” by the inductor.

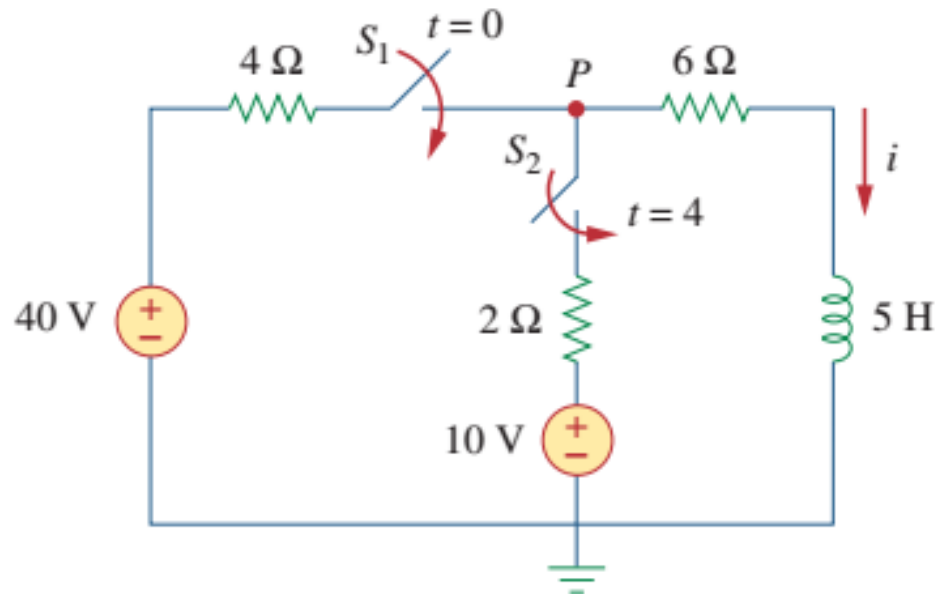


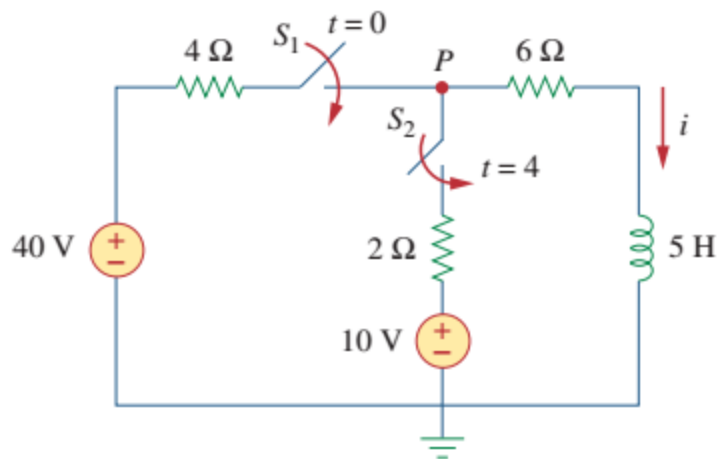
Find the time constant for the RC circuit



Sequential switch

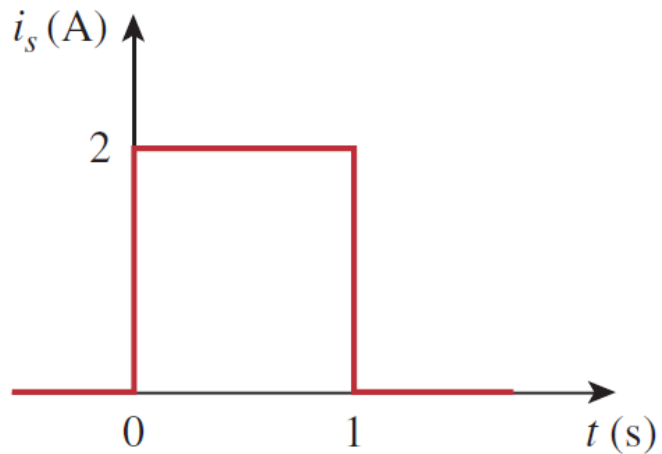
At $t = 0$, switch 1 in Fig. 7.53 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for $t > 0$. Calculate i for $t = 2$ s and $t = 5$ s.



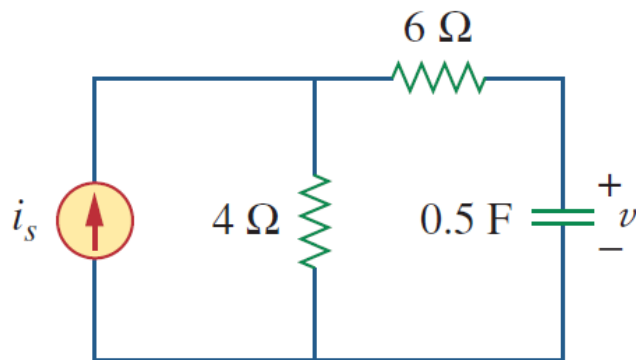




If the waveform in Fig. 7.116(a) is applied to the circuit of Fig. 7.116(b), find $v(t)$. Assume $v(0) = 0$.



(a)

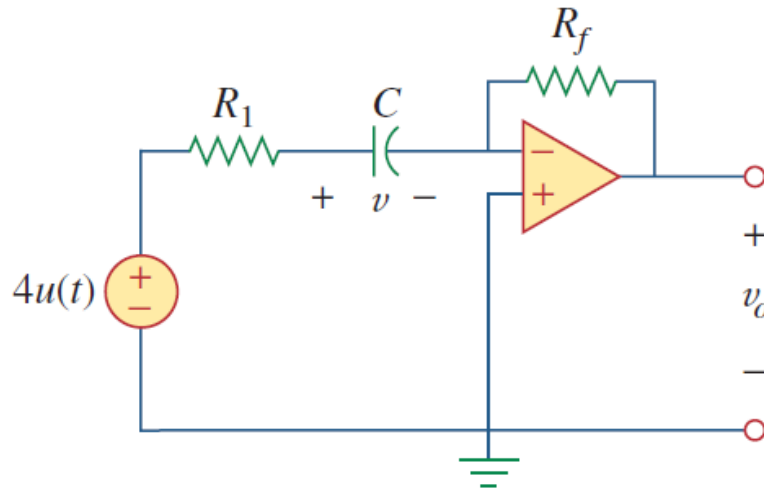


(b)



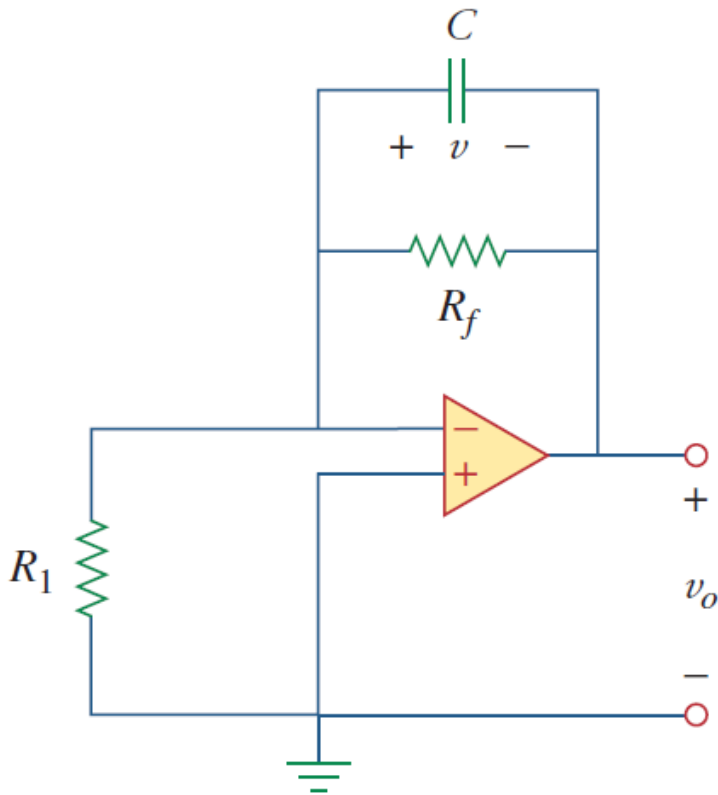
1st Order Circuit with OPA

For the op amp circuit of Fig. 7.138, let $R_1 = 10 \text{ k}\Omega$, $R_f = 20 \text{ k}\Omega$, $C = 20 \mu\text{F}$, and $v(0) = 1 \text{ V}$. Find v_o .



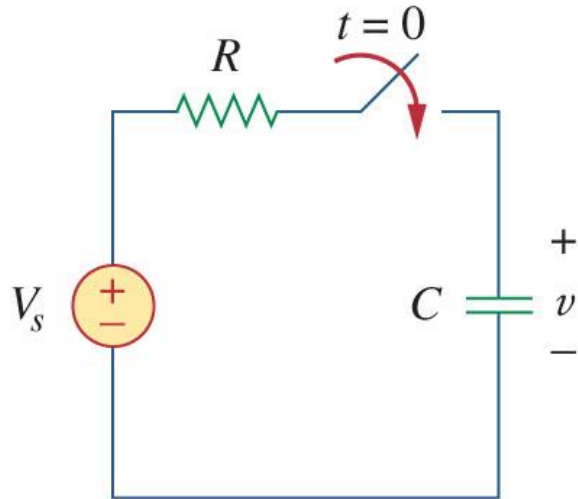


For the op amp circuit in Fig. 7.56, find v_o for $t > 0$ if $v(0) = 4$ V.
Assume that $R_f = 50$ k Ω , $R_1 = 10$ k Ω , and $C = 10$ μ F.



Other kinds of excitations

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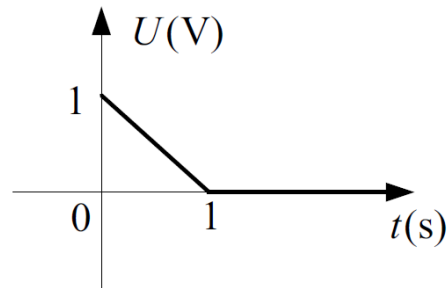
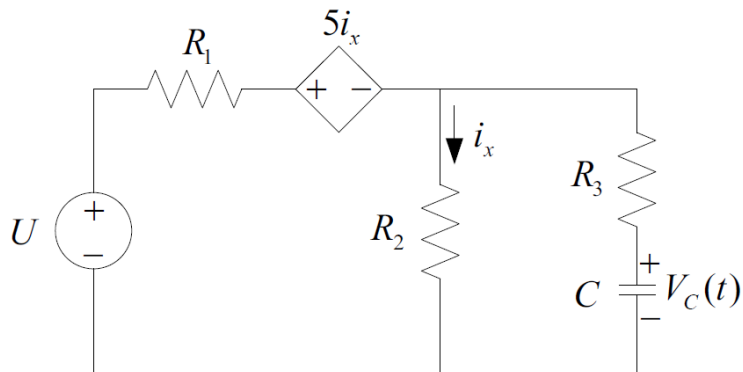
$$V_s = e^{\omega t}, \quad \omega t, \quad V_s = \cos(\omega t)$$

$$v(0^-) = v(0^+) = v_0$$



Other kinds of excitations

In the circuit below, $R_1 = 10\ \Omega$, $R_2 = 5\ \Omega$, $R_3 = 10\ \Omega$, $C = 10\ \text{mF}$. When $t < 0$, the input voltage (U) is $1\ \text{V}$. When $t = 0$, the input voltage begins to change as shown in the plot below. When $t > 1\ \text{s}$, $U = 0$. Assume that the circuit reaches steady state before $t = 0$. Determine the expression for $V_C(t)$ when $t \geq 0$.





Applications

